construction that may or may not have anything to do with ordinary space. It may have any number of dimensions from 1 to ∞ and it may have components that are integers, real numbers, or even more general things.

The vector spaces we use to define quantum mechanical states are called *Hilbert spaces*. We won't give the mathematical definition here, but you may as well add this term to your vocabulary. When you come across the term *Hilbert space* in quantum mechanics, it refers to the space of states. A Hilbert space may have either a finite or an infinite number of dimensions.

In quantum mechanics, a vector space is composed of elements |*A* > called *ket-vectors* or just *kets*. Here are the axioms we will use to define the vector space of states of a quantum system (*z* and *w* are complex numbers):

1. The sum of any two ket-vectors is also a ket-vector:

$$|A\rangle + |B\rangle = |C\rangle.$$

2. Vector addition is commutative:

$$|A\rangle + |B\rangle = |B\rangle + |A\rangle.$$

3. Vector addition is associative:

$$\{|A\rangle + |B\rangle\} + |C\rangle = |A\rangle + \{|B\rangle + |C\rangle\}.$$

4. There is a unique vector 0 such that when you add it to any ket, it gives the same ket back:

$$|A\rangle + 0 = |A\rangle.$$

5. Given any ket |*A*>, there is a unique ket −|*A*> such that